Does AI-Assisted Writing Resurrect Retracted Science? A Population-Level Study of Zombie Citations

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Abstract

Large language models trained before a retraction cannot know about it, and chatbot-level audits have shown that conversational models draw on retracted articles in a majority of relevant answers while flagging the retraction less than half the time. Whether this failure reaches the published literature is unknown. We measure it, combining the Retraction Watch database with a frozen OpenAlex snapshot to track citations to already-retracted papers, which we call zombie citations, across 204.9 million citing works from 2018 to 2026. Our design has three prongs, each informative on its own: an interrupted time series of the population zombie-citation rate around the release of ChatGPT, a matched-cohort comparison of 1,731 papers containing verbatim LLM artifacts against controls matched on venue, year, and citedness, and partial-identification bounds that convert a noisy lexical measure of AI involvement into an honest interval rather than an attenuated point estimate. The three prongs disagree in a way that is itself the finding. The population rate rises by 0.232 zombie citations per 10,000 references at the ChatGPT release, a 20% increase over the pre-period rate of 1.160, exceeding all 36 placebo breaks and concentrated in computational fields. Yet papers whose LLM involvement is certain cite retracted work no more often than their matched controls, at a ratio of 0.878 with a 95% interval of 0.379 to 1.632, while the lexical proxy appears to separate the literature roughly hundredfold and turns out to be measuring article length. Flagged papers carry 44.7 references on average against 9.2, and the outcome is a per-paper indicator, so the contrast is composition. Fitting the outcome on the proxy and log reference count together leaves reference count overwhelming (odds ratio 2.21 per log unit, $p = 2 \times 10^{-38}$) and the proxy undetectable (odds ratio 1.35, 95% interval 0.95 to 1.92). Neither observed nor inferred exposure predicts citing retracted work, so the population shift stands unattributed, and the proxies that would attribute it do not survive the obvious control. Data, code, and the artifact corpus are available at `https://github.com/garyzhang1006/ai-science-zombie-data`.

1 Introduction

Retraction is the scientific record's error-correction mechanism, and it works only if authors check the current status of what they cite. Large language models complicate that mechanism in a specific, foreseeable way: a model whose training data was collected before a retraction has no representation of it, and a model asked to summarize a field will reproduce the retracted claim with the same fluency as any other. The chatbot-level evidence is already troubling, since a recent audit found that ChatGPT drew on 61.5% of retracted stem-cell articles when answering domain questions and

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

mentioned the retraction in only 41.7% of those responses (Zhang et al., 2025), and a broader study reached similar conclusions across fields (Thelwall et al., 2025). What no study has measured is whether the published literature itself now cites retracted work at an elevated rate because of AI-assisted writing. That is the question this paper answers.

The measurement problem is harder than it looks, and the difficulty shapes our design. Citations to retracted papers are rare, AI involvement is observed only through noisy proxies, and a naive regression of zombie-citation rates on a lexical AI score is attenuated by proxy error, so a null from that regression would describe the proxy rather than the literature. We therefore built the study around three prongs whose failure modes do not overlap: a population interrupted time series that needs no per-paper attribution at all; a cohort in which AI involvement is certain, because the papers carry verbatim LLM artifacts such as "as an AI language model" that reach print only through unedited pasting; and set-identified bounds that make the lexical score of Kobak et al. (2024) and Liang et al. (2025) honest instead of discarding it, in the tradition of partial identification under misclassification (Manski, 2003).

Running all three was worth the cost, because they disagree. The population series moves at the release of ChatGPT. The cohort with certain exposure does not move at all. The lexical proxy separates sharply until reference count is controlled, and then not at all. A study built on any one prong would have reported a clean result and been wrong about what it measured.

In the same cohorts we track a second, denser outcome: references that do not resolve, which prior evidence links to model fabrication far above the human baseline (Walters and Wilder, 2023; Bhattacharyya et al., 2023). A zombie citation is real-but-refuted knowledge propagating; an unresolvable reference is fabricated support.

Our contributions are: (1) the first population-level measurement of zombie-citation dynamics across the LLM transition, covering 204.9 million citing works and 216,514 zombie citation events; (2) the artifact cohort, a public corpus of 1,731 papers with certain LLM involvement and matched controls, together with a measured false-positive rate for every artifact phrase we use; (3) bounded rather than attenuated estimates of the dose-response between lexical AI involvement and citation integrity, computed within strata of reference-list length; and (4) a demonstration that this proxy's apparent association with citation integrity is entirely explained by reference count, which is a warning about a measurement now in wide use.

2 Related work

Post-retraction citation is a longstanding integrity concern predating language models, covering the persistence of citations to retracted work and interventions such as the RISRS recommendations (Schneider et al., 2022). Case studies have traced the citation network of a single prominent retracted paper and found direct citation to be the main propagation channel (van der Vet and Nijveen, 2016). We add scale and a treatment contrast, measuring the phenomenon across the full indexed literature and asking whether the LLM transition changed its rate.

On the AI side, audits of conversational models document both the retraction blindness described above (Zhang et al., 2025; Thelwall et al., 2025) and the fabrication of bibliographic references, which Walters and Wilder measured at 55% of GPT-3.5 citations and 18% of GPT-4 citations, with substantive errors in a further 24% of the GPT-4 citations that do refer to real work (Walters and Wilder, 2023; Bhattacharyya et al., 2023). Verbatim artifact phrases in published papers were catalogued by Strzelecki across premier journals (Strzelecki, 2025), establishing that unedited model output does reach print; we industrialize that observation into a matched-cohort design. Population-level prevalence estimation (Kobak et al., 2024; Liang et al., 2025; Thelwall, 2025) provides our dose proxy, and its error rates feed the bounds. To our knowledge no prior work connects these threads to measured citation-integrity outcomes in the literature itself.

3 Data and design

3.1 Retraction and citation data

Retraction events come from the Retraction Watch database, which yields 60,247 retractions carrying both a usable DOI and a retraction date, of which 59,924 (99.5%) match a work in OpenAlex.

Table 1: Artifact phrase registry with false-positive rates measured against the pre-ChatGPT era. A phrase appearing in a paper published before 2023 cannot be unedited model output, so the pre-2023 share of hits estimates the phrase's false-positive rate.

Phrase	Hits	Pre-2023	FP rate	Status
as an AI language model	9,131	111	1.2%	retained
as a large language model, I	244	1	0.4%	retained
as of my last knowledge update	157	3	1.9%	retained
as an AI developed by OpenAI	99	3	3.0%	retained
my knowledge cutoff	89	0	0.0%	retained
I don't have access to real-time	22	1	4.5%	retained
I cannot browse the internet	3	0	0.0%	retained
Regenerate response	9,085	7,926	87.2%	excluded

Citation events and reference lists come from the OpenAlex parquet snapshot of 26 June 2026, pinned to that single release so that every number here is reproducible against a frozen corpus rather than a moving index. We work from the snapshot rather than the REST API by necessity: the API now meters usage at 1,000 calls or $0.10 per day on its free tier, which cannot cover a scan of half a billion works, and the snapshot is unmetered.

A citation is a zombie citation if the citing paper appeared at least six months after the cited paper's retraction date, the buffer absorbing indexing lag, and we exclude citing works that mention retraction in their title or abstract, since citing a retracted paper to discuss its retraction is the correction system working as intended. Scanning all 2,446 snapshot files yields 216,514 zombie citation events across 204.9 million citing works published between 2018 and 2026. Both the numerator and the denominator come from the same pass, so the monthly rate is exact rather than estimated from a sample.

3.2 The artifact cohort

Candidate artifact papers are harvested by full-text search over a registry of verbatim LLM phrases. This step needs more care than it first appears, and getting it wrong would have destroyed the cohort. OpenAlex full-text search is stemmed, so the phrase "regenerate response" matches "regenerative response" and returns 9,085 hits dominated by zebrafish and stem-cell regeneration papers. Those hits would have made up 82% of our cohort.

We caught this with a negative control that the study period supplies for free. No paper published before 2023 can contain unedited ChatGPT output, so every pre-2023 hit is a false positive by construction, and the pre-2023 share of a phrase's hits estimates its false-positive rate directly. Table 1 reports that rate for every phrase we screened. "Regenerate response" fails at 87.2% and is excluded. The seven surviving phrases range from 0.0% to 4.5%, with the workhorse phrase "as an AI language model" at 1.2% across 9,131 hits. This converts the claim that these phrases make model involvement certain from an assertion into a measurement.

Filtering then removes papers whose own subject is language models, since those quote the phrases deliberately: 3,613 candidates are dropped on title, 1,868 on subfield, and 1,549 for carrying no references. What remains is 1,731 papers, all published in 2023 or later, of which 1,643 carry a venue identifier. Each is matched to five controls from the same venue and year, within 25% of its reference-list length and nearest in citation count, giving 7,644 matched pairs. Reference-list lengths come out nearly identical across arms, 42.55 against 42.45, so the matching binds. Within this cohort the exposure is observed rather than proxied, at the price of representing only the least careful tail of AI-assisted writing, a selection we return to in Section 6.

3.3 Outcomes, series, and bounds

For every paper we compute two outcomes over its reference list: the count of zombie citations as defined above, and the count of unresolvable references, meaning entries whose DOI fails to resolve in Crossref and whose bibliographic string matches no indexed work under fuzzy search at a similarity threshold of 85. The population series aggregates the first outcome monthly within

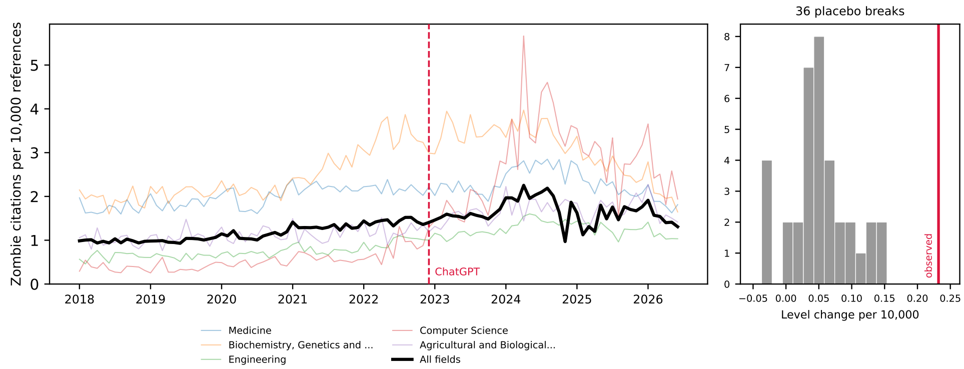

Figure 1: Zombie citations per 10,000 references, monthly, 2018–2026, for the five fields contributing the most events and for the literature as a whole. The vertical line marks the ChatGPT release; the inset shows the distribution of 36 placebo break estimates from 2019–2021 against the estimated 2022 break.

field strata, and the interrupted time series allows a level and slope change at the ChatGPT release in December 2022, with Newey-West standard errors and placebo breaks at each month of 2019 through 2021 calibrating the procedure's own false-positive rate.

The bounding analysis follows Manski (2003). Given a lexical AI score with assumed sensitivity and specificity, the observed outcome-by-score table set-identifies the true outcome rates among AI-assisted and unassisted papers. We solve this over a grid rather than at any single assumed point, and the grid is disciplined by the data: our rule flags 0.86% of 2023-onward papers, and the prevalence solution requires specificity above $1 - P(S{=}1) = 0.991$, so any specificity below that admits more false positives than observed positives and no solution at all. A rule demanding two distinct markers from a 37-term list is high-specificity and low-sensitivity by construction, and the grid sits in that regime.

The outcome for this prong is a per-paper indicator, whether a paper carries at least one zombie citation, and that indicator rises with the length of a paper's reference list. The lexical rule needs two markers inside an abstract, so it selects papers with long abstracts, which are full research articles carrying many references. Across the unrestricted sample, flagged papers average 44.7 references against 9.2 for the rest. We therefore compute the bounds within reference-count strata, and take papers with at least 20 references as the primary specification, which compares full articles to full articles at 61.2 references against 50.8.

4 Results

4.1 Population dynamics

The population rate of zombie citation runs at 1.160 per 10,000 references over the pre-period and rises through it, and the interrupted time series estimates a level change of $+0.232$ at the ChatGPT release (standard error 0.129, $p = 0.074$), a 20% increase on the pre-period rate. The conventional p-value is marginal, but it is the weaker of the two tests available here. Across 36 placebo breaks placed through 2019 to 2021, the largest absolute level change is 0.154, so the real break exceeds every placebo the procedure produces on data where nothing happened, a placebo-based p below $1/37$ (Figure 1).

The level change is stable under truncation: refitting with the series ending in June 2025, December 2025, March 2026, and June 2026 gives $+0.193$, $+0.190$, $+0.194$, and $+0.232$. The slope change is not. It reads -0.0116 ($p = 0.005$) on the full series but decays to -0.0086 ($p = 0.312$) when the last year is dropped, and the snapshot's final months are visibly under-indexed, carrying 9.0 million references in June 2026 against a monthly norm near 15 million. We therefore report the level shift as the population result and treat the post-break decline as an artifact of the indexing tail.

Table 2: Artifact cohort against matched controls. Rates per 1,000 references; ratios with 95% cluster-bootstrap intervals resampling papers. The unresolvable-reference row rests on 7,873 papers and 415,750 references checked against Crossref.

Outcome	Artifact papers	Matched controls	Ratio
Zombie citations	0.157	0.179	0.878 [0.379, 1.632]
Unresolvable references	138.84	127.74	1.087 [1.005, 1.174]
Reference-list length	42.55	42.45	

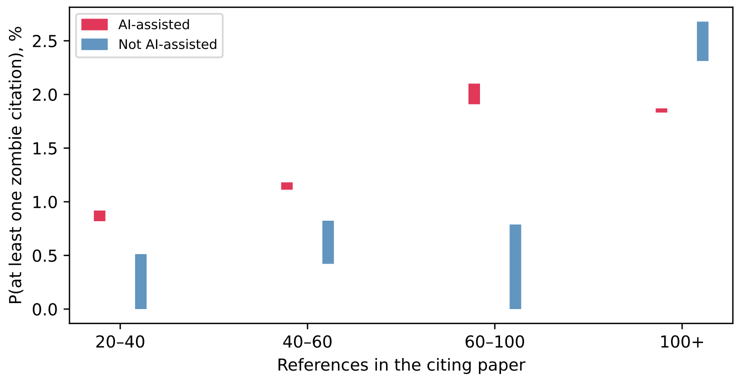

Figure 2: Set-identified probability of carrying at least one zombie citation, by lexical AI score, within strata of reference-list length. Bars span the full range of solutions across the grid. Each flagged bar rests on 6 to 11 events, so the strata illustrate the direction of the adjustment rather than establishing a dose-response.

The break concentrates where one would expect AI writing tools to have landed first. Computer Science shows a level change of $+1.489$ ($p = 0.006$), against $+0.606$ in Biochemistry ($p = 0.043$), $+0.296$ in Engineering ($p = 0.010$), $+0.297$ in Agricultural and Biological Sciences ($p = 0.005$), and $+0.174$ in Materials Science ($p = 0.017$). Medicine, the largest field by volume, does not move ($+0.163$, $p = 0.362$). Eight fields were tested, and Computer Science ($p = 0.0057$) and Agricultural and Biological Sciences ($p = 0.0045$) are the two that survive a Bonferroni threshold of 0.00625; we read the rest as suggestive of a computational concentration rather than as separately established.

4.2 The artifact cohort

Within the cohort where LLM involvement is certain, the zombie-citation result is a null. Artifact papers cite retracted work at 0.157 per 1,000 references against 0.179 among controls, a ratio of 0.878 whose 95% interval, 0.379 to 1.632, comfortably contains parity (Table 2). This is a bounded null rather than an absence of evidence: the design rules out effects above roughly 1.6 times the control rate, and the point estimate sits below one. The endpoint is simply sparse. At 0.157 per 1,000 references, 1,645 artifact papers carrying 42.55 references each are expected to produce about eleven zombie citations in total, and no cohort of this size can resolve a modest effect on such a rate.

The denser endpoint carries what statistical weight the cohort has, as the design anticipated. Artifact papers carry 138.84 unresolvable references per 1,000 against 127.74 among controls, a ratio of 1.087 with an interval of 1.005 to 1.174. The effect is real but small, and its interval touches parity closely enough that we decline to describe it as robust. Its absolute level also deserves care: an unresolvable rate near 13% across both arms is far too high to represent fabrication, and mostly reflects Crossref indexing gaps for books, theses, and items without DOIs. Only the ratio between matched arms is interpretable, and only under the assumption that those indexing gaps fall symmetrically across artifact and control papers drawn from the same venue and year.

4.3 Bounded dose-response

Applied without restriction to a seeded sample of 629,900 papers published from 2023 onward, this prong produces the largest contrast in the study and a spurious one. The lexical rule flags 0.86% of papers. Among those, 1.057% carry at least one zombie citation against 0.113% among the rest, and the correction set-identifies the flagged rate within [1.24%, 5.11%] against [0.00%, 0.089%], intervals that do not overlap anywhere on the grid.

That comparison is about article type. Flagged papers average 44.7 references against 9.2, because a rule requiring two markers inside an abstract selects long-abstract full articles, and the outcome counts whether a paper carries any zombie citation at all. A paper with sixty references has more opportunities to carry one than a paper with six, whoever wrote it.

Restricting to papers with at least 20 references brings the arms to 61.2 references against 50.8, and the naive contrast falls from roughly ninefold to 1.55 (1.261% against 0.816%). Stratifying further leaves flagged papers higher in three of four strata and lower in the fourth (Figure 2), but those cells rest on 6 to 11 flagged events each, and the apparent reversal above 100 references is 6 events against 105 with a Fisher p of 0.84. We report the strata because the shrinkage across them is informative, and we decline to read a dose-response into them.

The test that settles the question fits the event indicator on the proxy and on log reference count together, across all 78,465 papers with at least 20 references and 652 events. Reference count dominates, at an odds ratio of 2.21 per log unit with $p = 2 \times 10^{-38}$. The proxy does not survive it: an odds ratio of 1.35 with a 95% interval of 0.95 to 1.92 and $p = 0.097$. On the full sample the proxy appears significant (odds ratio 1.54, $p = 0.008$) only because reference count there ranges from 1 to several hundred and carries an odds ratio of 3.71. Once the mechanical advantage of a long bibliography is removed, the lexical AI score carries no detectable information about whether a paper cites retracted work.

This is the paper's most transferable result. Excess-vocabulary proxies are already used to compare prevalence across subpopulations: Kobak et al. (2024) report a lower bound that "differed across disciplines, countries, and journals, reaching 40% for some subcorpora." Any such comparison inherits whatever else the markers track, and pairing those estimates with an outcome inherits it twice. Here the markers track abstract length, and through it reference count, strongly enough to turn a null into a hundredfold separation. Misclassification bounds do not protect against this, since the correction assumes the proxy errs at random with respect to the outcome, and length-driven selection violates that in the direction that inflates the result.

5 Discussion

The three prongs give three different answers, and the disagreement is the paper's substance. The population rate moves at the ChatGPT release by an amount that survives every placebo the data can generate. Papers whose LLM involvement is certain show nothing on the retracted-citation endpoint and a small effect on the denser one. The lexical proxy appears to separate sharply and carries no detectable signal once reference count is controlled.

The three now agree on the per-paper question: neither observed nor inferred exposure predicts citing retracted work, and the design that appeared to disagree was measuring article length. The cohort where exposure is observed rather than inferred is where a causal effect should be largest, and it is absent there. What the proxy separates is a population that differs from the rest of the literature in more ways than its use of language models, most obviously in how much it cites.

This leaves the population result standing and unexplained, which we think is the honest place to leave it. The break is concentrated in computational fields, survives 36 placebo breaks and four series endpoints, and coincides with the arrival of AI writing tools. It is equally consistent with contemporaneous changes in indexing, in citation practice, or in who was publishing in those fields in 2023. Distinguishing these needs per-paper exposure that is neither self-reported nor lexical, which is a real gap we cannot close with public data.

The artifact-cohort null needs one qualification. The cohort is selected on carelessness, since a paper that ships "as an AI language model" verbatim received no meaningful proofreading, so its null bounds the harm of the careless tail rather than of AI assistance in general. That tail is also where integrity screening would go first, which makes the null directly useful to anyone planning such screening.

For the workshop's framing question, the study offers a measured instance of a general worry and a sharper caution about how the worry gets measured. The aggregate rate at which the literature cites retracted work did shift when AI writing tools arrived. Whether AI assistance caused it remains unresolved by our strongest design, and the lexical proxies now in wide circulation would have answered the question confidently and, on this evidence, wrongly.

6 Limitations

The unresolvable-reference comparison queried every one of the 9,009 intended papers, of which 7,873 (87.3%) deposit reference data in Crossref at all; the remainder are absent from the comparison for a reason unrelated to their content. OpenAlex full-text coverage is partial and skewed toward open access, so the artifact cohort undercounts paywalled carelessness. The artifact filter's precision is established by the era-based false-positive rates in Table 1 rather than by manual adjudication, which we did not perform; the era test bounds how often a phrase fires on text no model could have written, and says nothing about papers where the phrase is quoted by a 2023-or-later author discussing model output, which the topic and title filters target instead. We publish the full candidate list with every drop reason, together with a seeded 100-paper sample, so that any reader can audit both. Retraction Watch coverage is itself uneven across fields and eras. The interrupted time series identifies a break at the ChatGPT release but cannot by itself attribute the break to LLM writing rather than to contemporaneous changes in indexing or citation practice, which the placebo distribution mitigates without eliminating. The bounds correct for misclassification of the AI proxy and not for confounding, and Section 4 shows that the confounding is large enough to dominate the unconditioned estimate; the conditioned estimate holds reference count fixed but nothing else, so field, venue, and author language remain uncontrolled. Finally, all three prongs measure citation to retracted work, and none measures whether the citing text endorses the retracted claim, so our rates are upper bounds on endorsement.

7 Conclusion

The literature's zombie-citation rate rose about 20% when AI writing tools arrived, against a pre-period rate of 1.160 per 10,000 references, and the papers we can prove were written with a language model show no such elevation. We cannot close that gap with public data, and we think saying so is more useful than picking whichever prong tells a cleaner story. The measurement lesson generalizes past this question: an excess-vocabulary proxy separated the literature roughly hundredfold on citation integrity, and the separation did not survive controlling for reference count. The artifact cohort with its measured per-phrase false-positive rates, the bounds machinery, and the monthly series are released at `https://github.com/garyzhang1006/ai-science-zombie-data`.

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A Artifact phrase registry and filtering

Table 1 gives the full registry with raw counts and measured false-positive rates. The filter drops candidates in this order, recording a reason for each: phrase retired for stem collision (9,088), published before 2023 (119), subject classified in an AI subfield (1,868), title matching a language-model topic pattern (3,613), abstract carrying two or more meta-discussion terms (125 across variants), document type outside the article, review, chapter, preprint, and dissertation set (approximately 500), and no reference list (1,549). The complete candidate table with per-paper decisions ships in the repository as `artifact_candidates.csv`, and a seeded 100-paper sample for hand validation ships as `hand_validation_sample.csv`.

B Robustness

The series is refit at four endpoints in Section 4; the level change varies between $+0.190$ and $+0.232$ while the slope change loses significance, which is why we report only the level shift. The zombie definition uses a six-month buffer after the retraction date; the buffer, the matching tolerance of 25%, and the fuzzy-match threshold of 85 are set in `config.py` and can be varied by rerunning the pipeline. The misclassification grid spans sensitivities of 0.03 to 0.20 and specificities of 0.992 to 0.999, and all sixteen combinations admit a solution; the identification constraint that fixes the lower edge of the specificity range is derived in Section 3. Bounds use reference-count strata, with at least 20 references as the primary cut; `bounds.json` reports the unrestricted fit, the primary fit, and every stratum.